

UNIVERSITY OF OSLO

A Self-Adaptive Digital Twin Architecture for Dynamic Resource Management

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SEAMS 2026

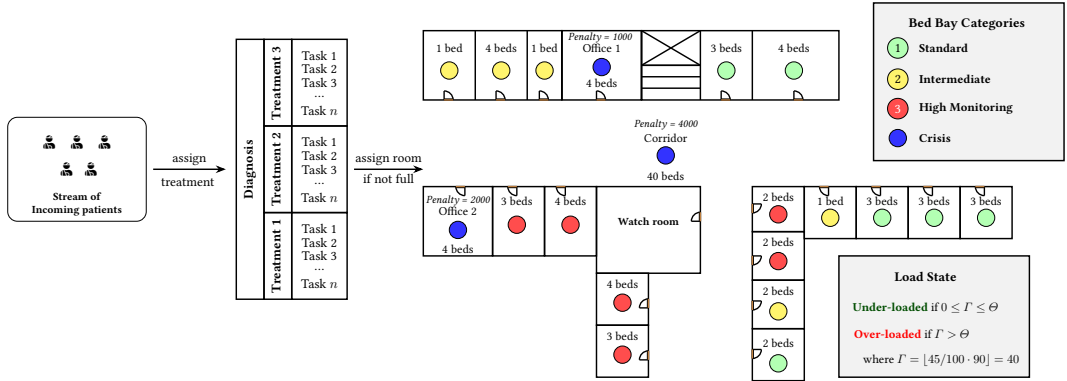
April 13, 2026



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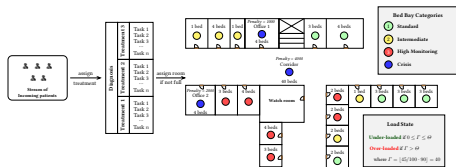
- ① Rationale
- ② Digital Twin Concepts
- ③ Digital Twin Architecture

Why do we need it?



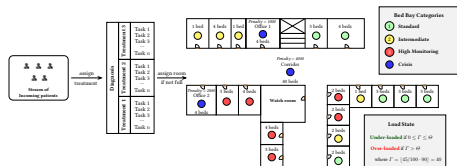
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- Hospital demand changes quickly; resource capacity does not



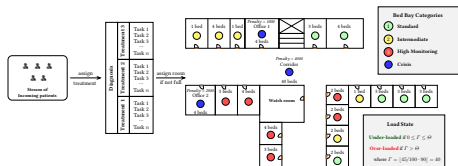
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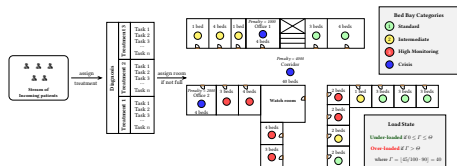
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Why do we need it?

- Hospital demand changes quickly; resource capacity does not
- Admission is often handled manually
- Depends on several factors
- Static architectures struggle when the underlying structure changes
- We need runtime adaptation with trustworthy model alignment



Baseline: BedreFlyt and the gap

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- BedreFlyt combines semantic modelling, forecasting, and optimization
- It produces bed-bay recommendations for human operators
- Human-in-the-loop actions can diverge from DT suggestions
- Main gap: the original ward layout is treated as fixed

What is the goal

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 - Penalty-based optimisation for cost-aware reconfiguration
- Human-in-the-loop supervision over the adaptation loop

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- 1 Rationale
- 2 Digital Twin Concepts
 - Modeling aspects
- 3 Digital Twin Architecture

What is a Digital Twin

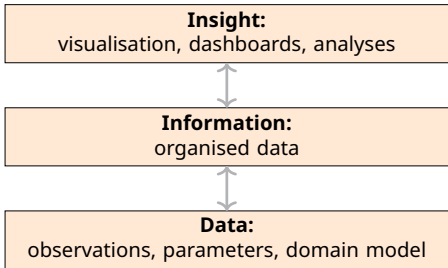
Digital twins are, as defined by NASEM

NASEM Definition of DT (2024)

A digital twin is a set of virtual information constructs that mimics the structure, context, and behaviour of a natural, engineered, or **social system** (or system-of-systems), is dynamically updated with data from its physical twin, has a predictive capability, and **informs decisions** that realise value. The bidirectional interaction between the virtual and the physical is central to the digital twin^a.

^aNational Academies of Sciences, Engineering, and Medicine (NASEM) (2024): Foundational Research Gaps and Future Directions for Digital Twins. The National Academies Press

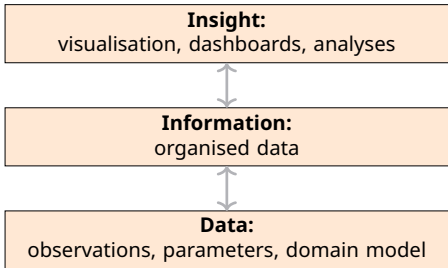
Digital Twins components



Capabilities for different insights:

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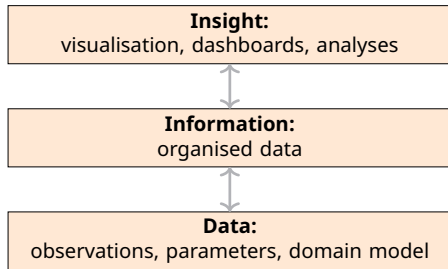
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Digital Twins components



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- **Prescriptive:** Advise on possible outcomes ("what if / what should")

Digital Twins components

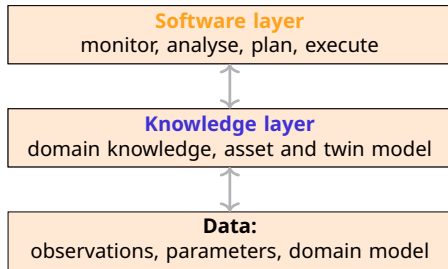
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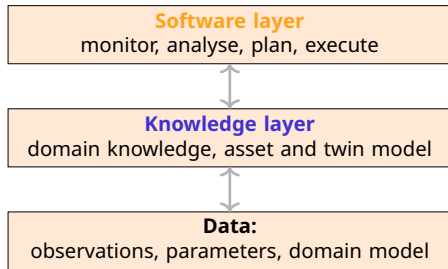
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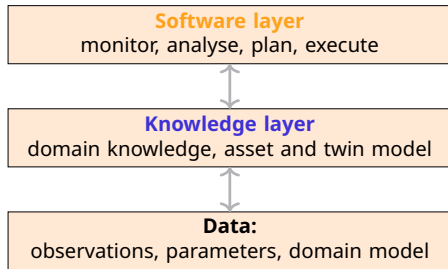
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- DT must adapt both structurally and behaviourally
- States of the DT cycles over time and they need to be captured

Twinning in SMOL

Programming support for twins with a behavioural and a structural layer
smolang.org



Twinning in SMOL

Programming support for twins with a behavioural and a structural layer

SMOL: Semantic Model Object Language

- Smalll OO programming system
- Ontology reasoners allow querying the KB
- Used to create the digital model

smolang.org



Twinning in SMOL

Programming support for twins with a behavioural and a structural layer
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behavioural twins in SMOL

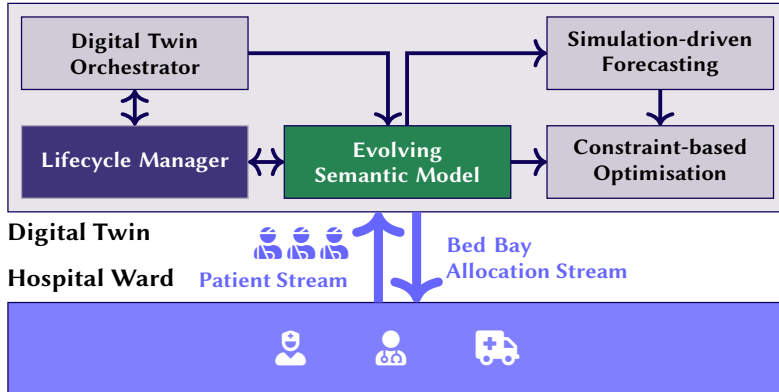
- SMOL can encapsulate (simulation) models based on the FMI standard
- Can automatically adapt the object states in the KB at runtime through semantic lifting



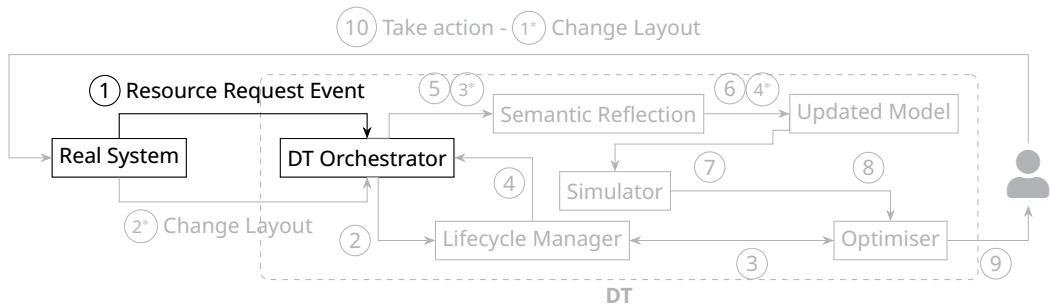
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 - Bedreflyt DT Execution

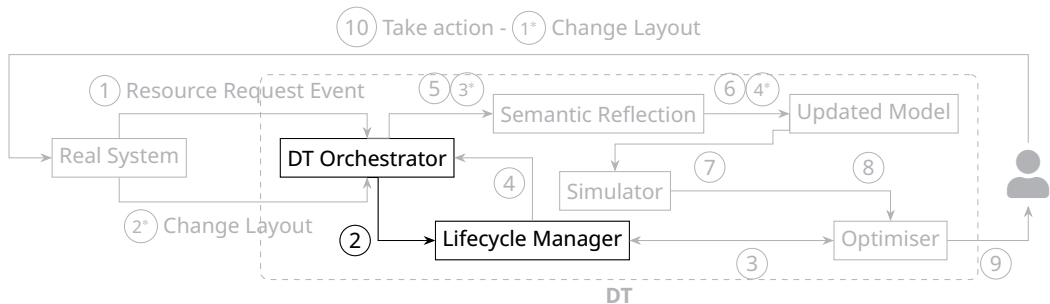
Architecture



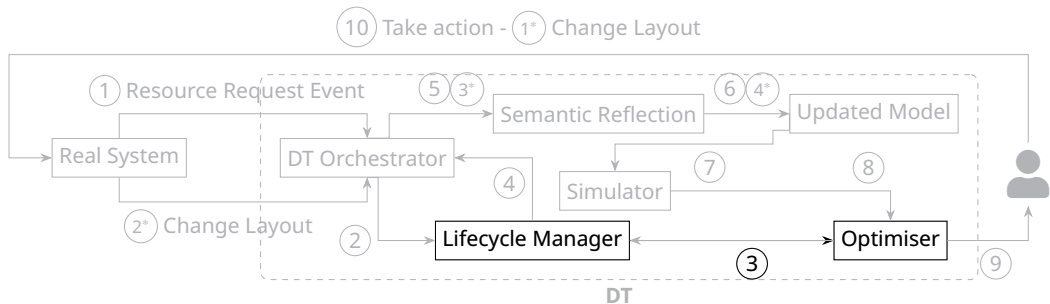
Human-in-the-loop



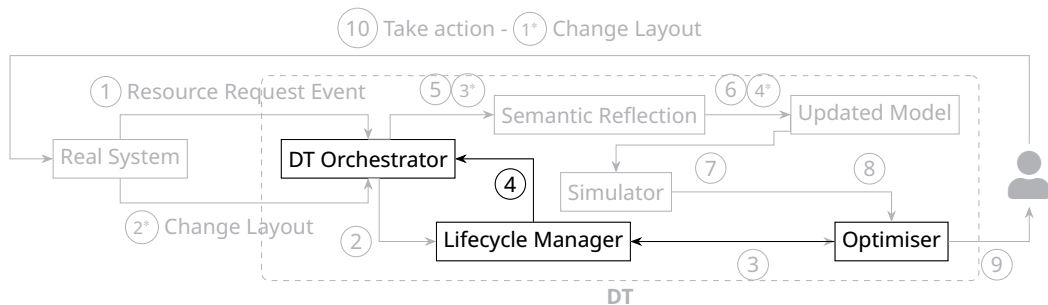
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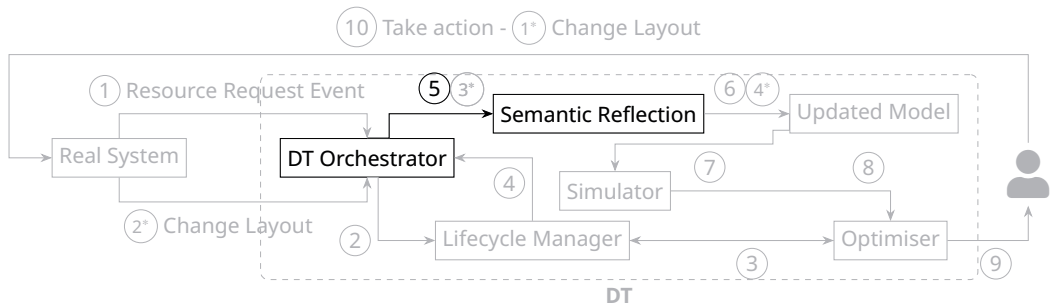
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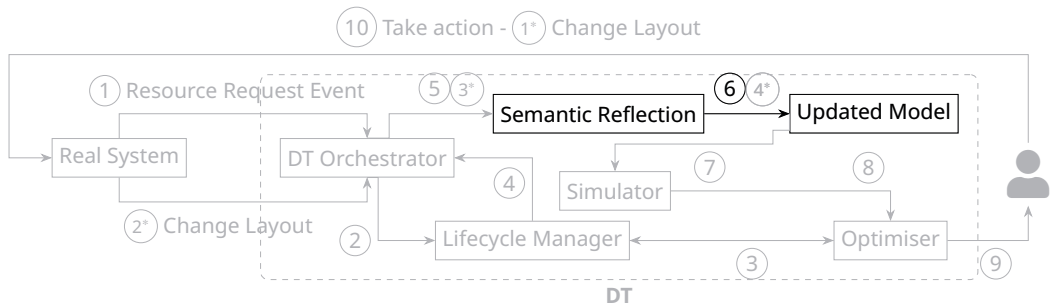
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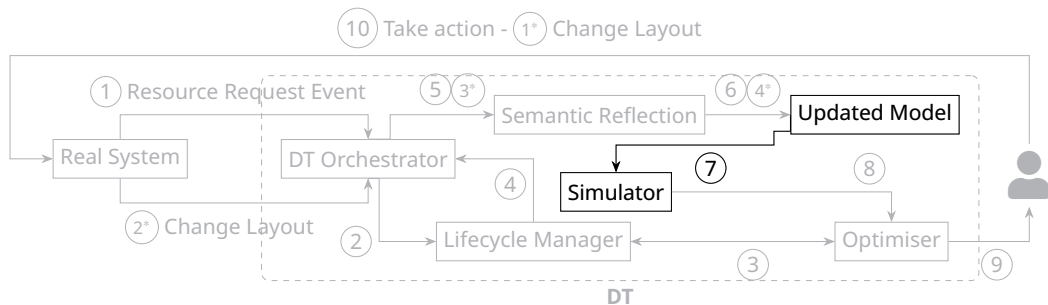
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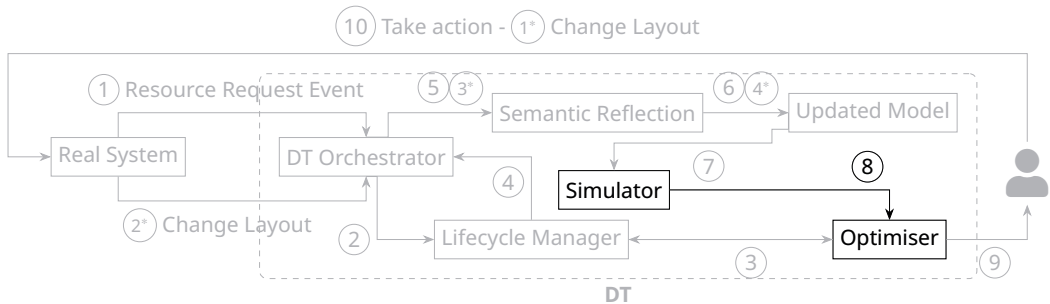
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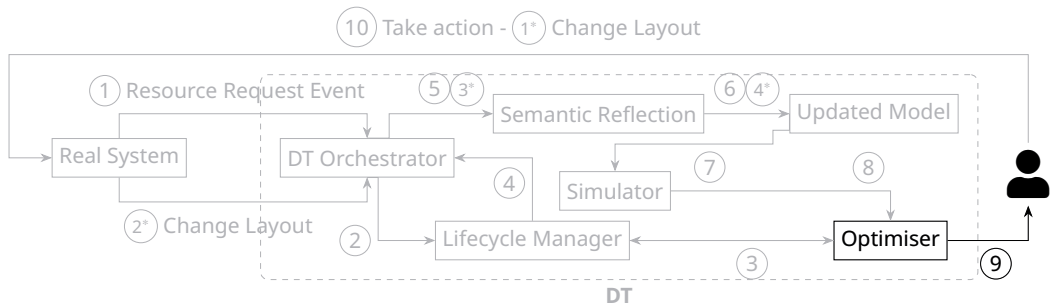
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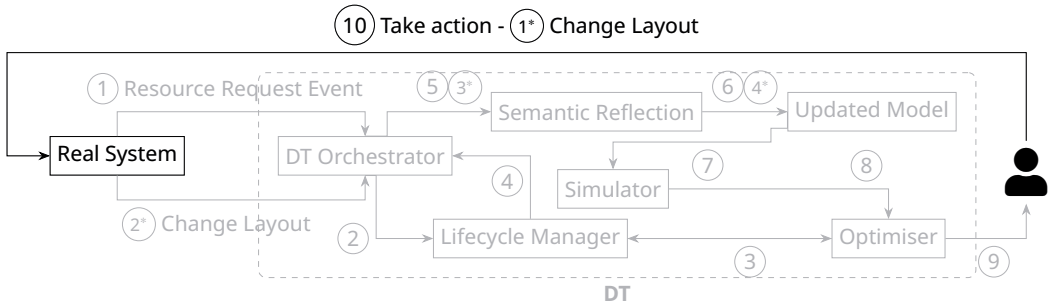
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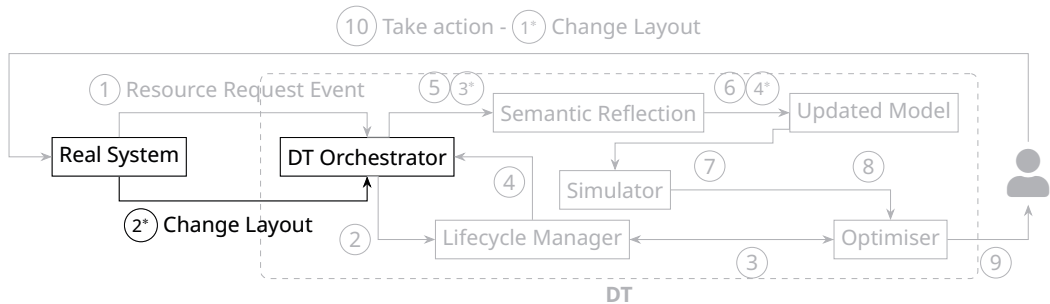
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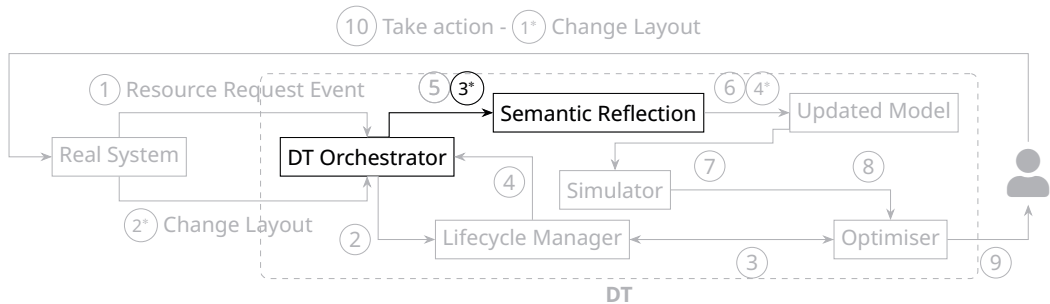
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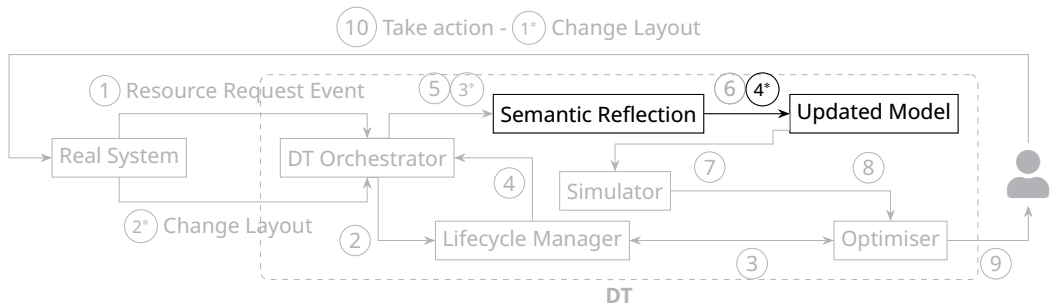
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Human-in-the-loop (cont.)

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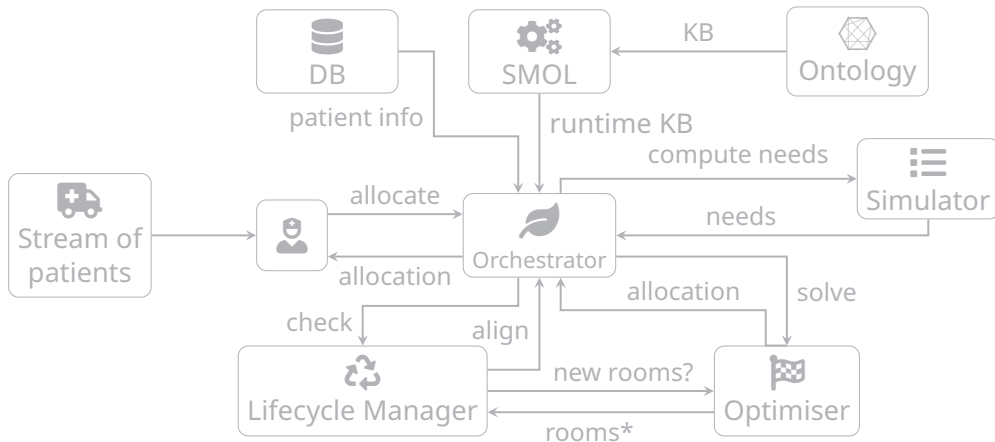
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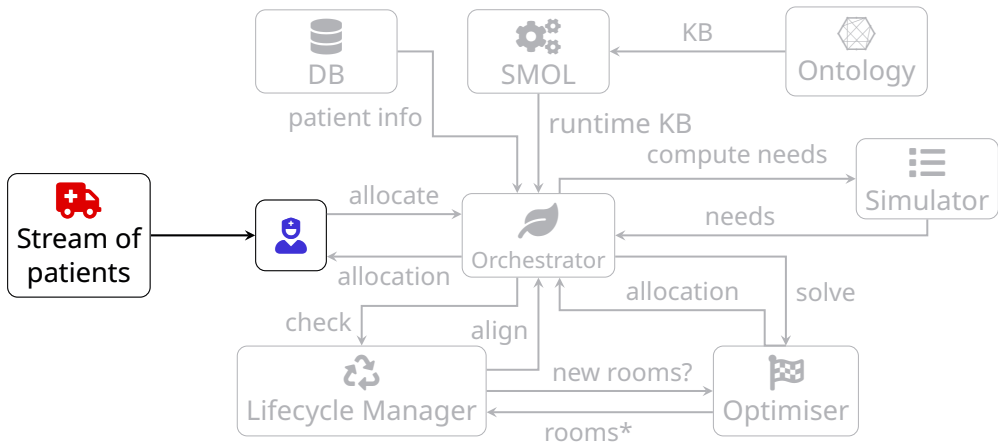
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- This enables supervised autonomy: controlled self-adaptation instead of static decision support

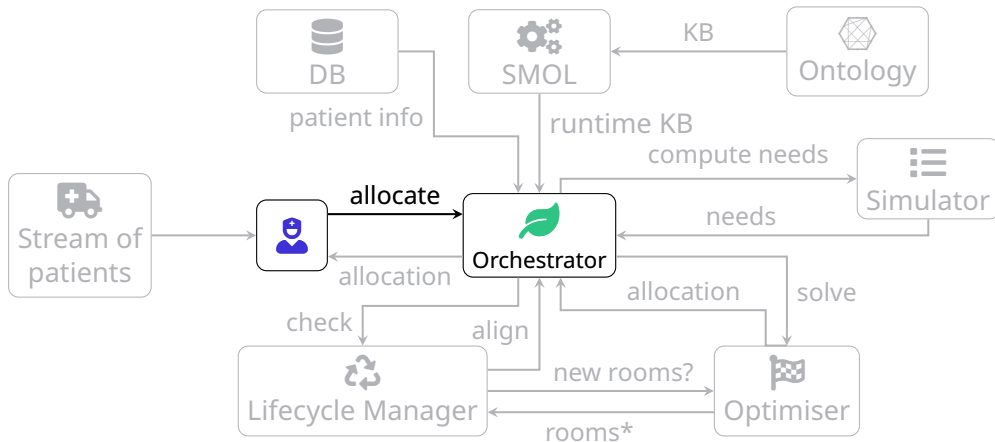
Allocation of patients



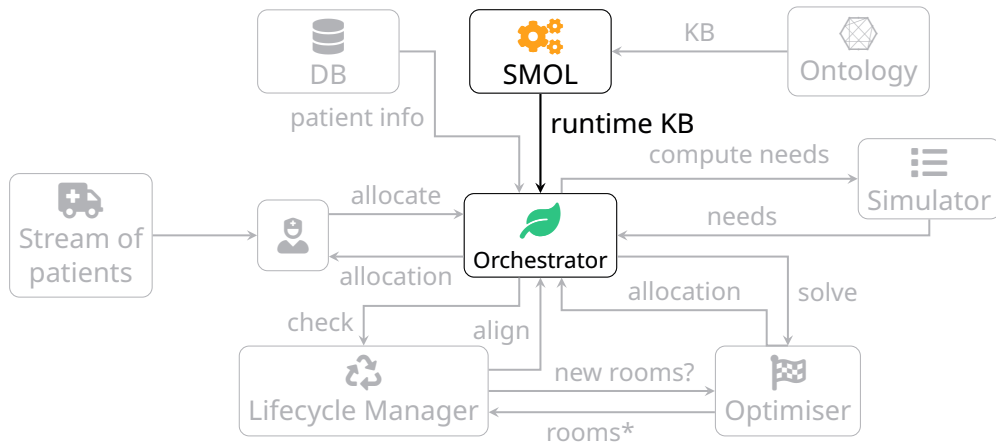
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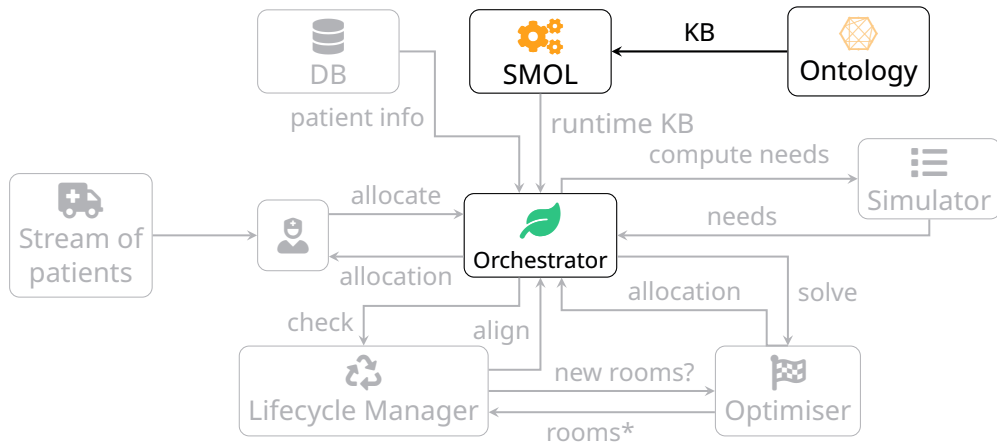
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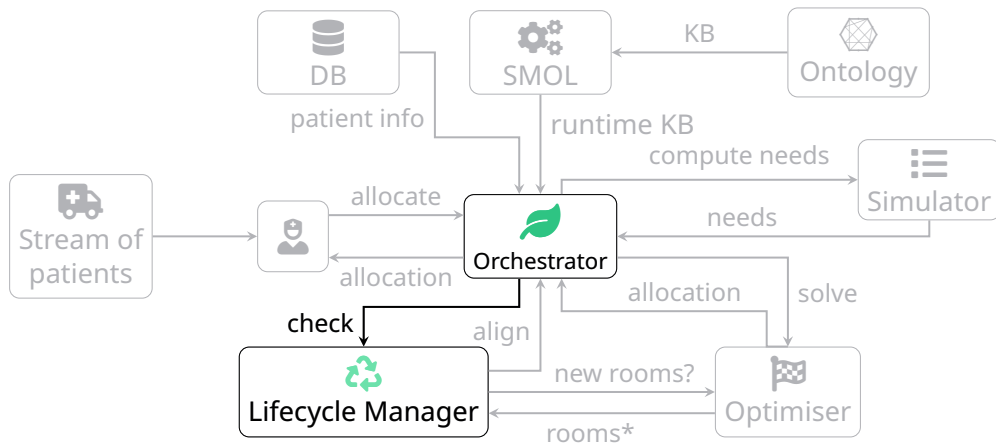
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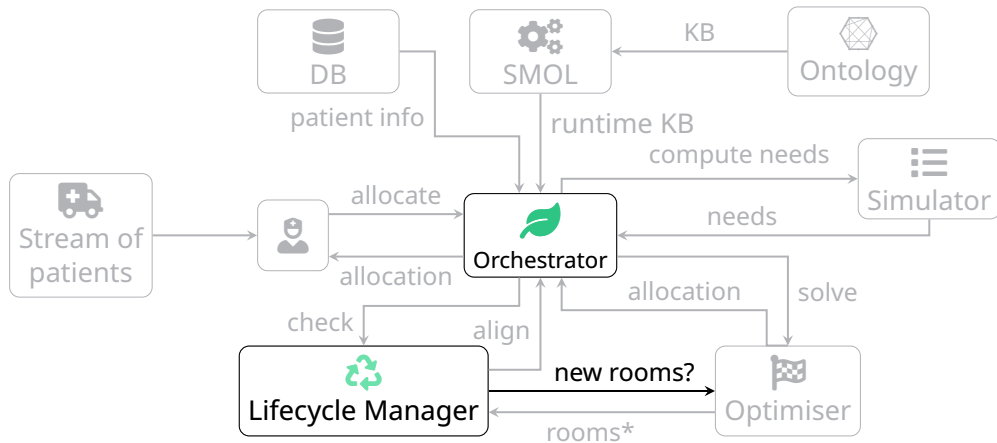
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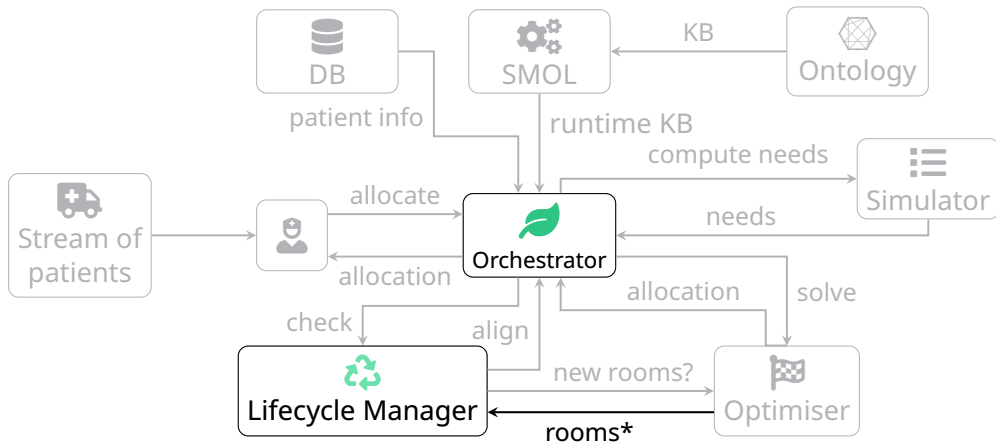
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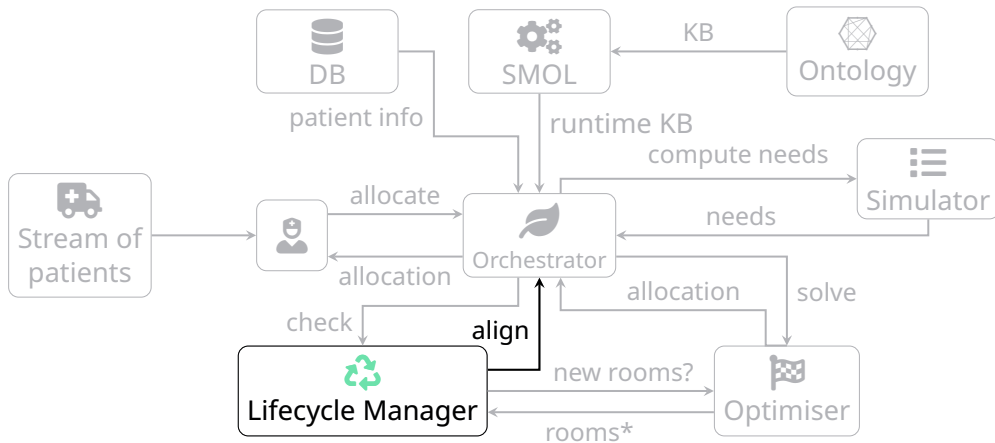
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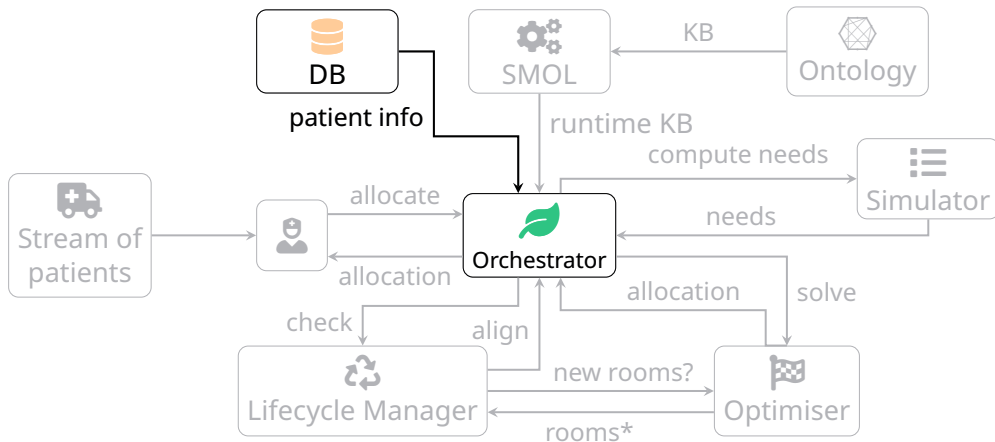
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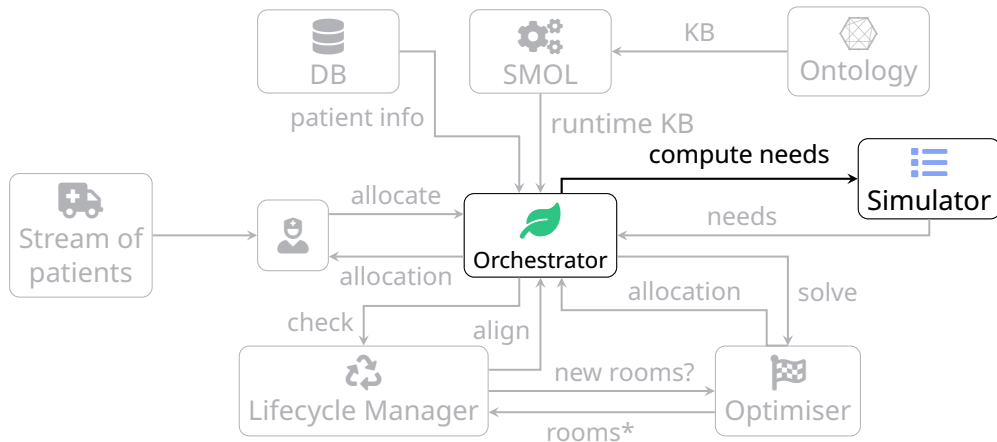
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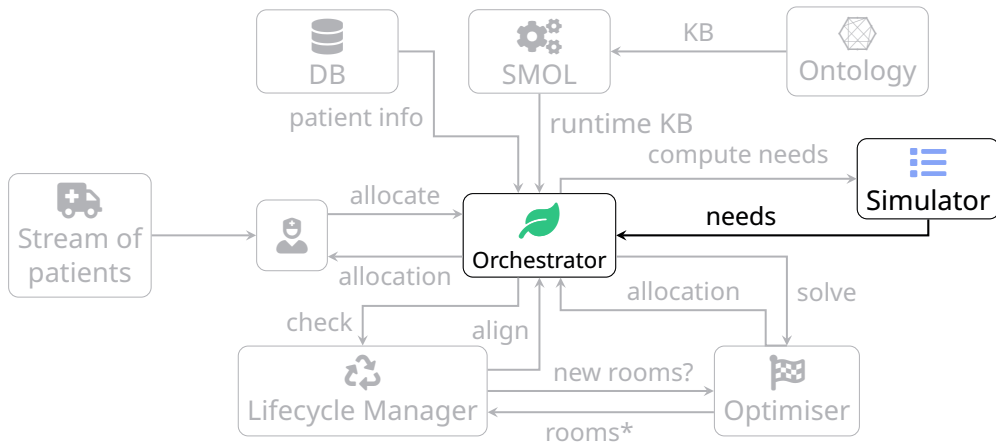
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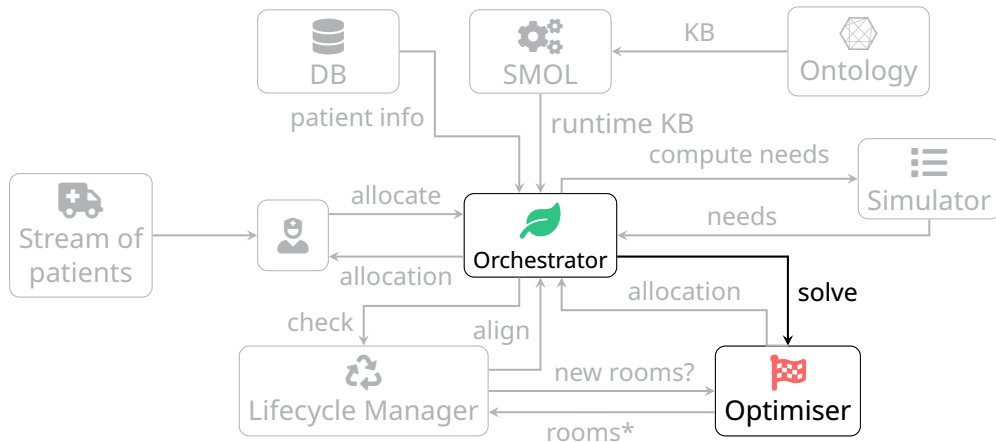
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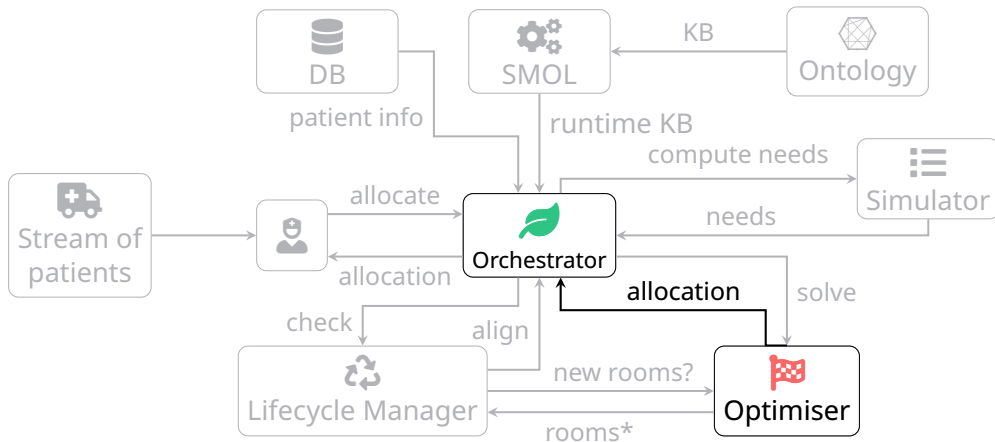
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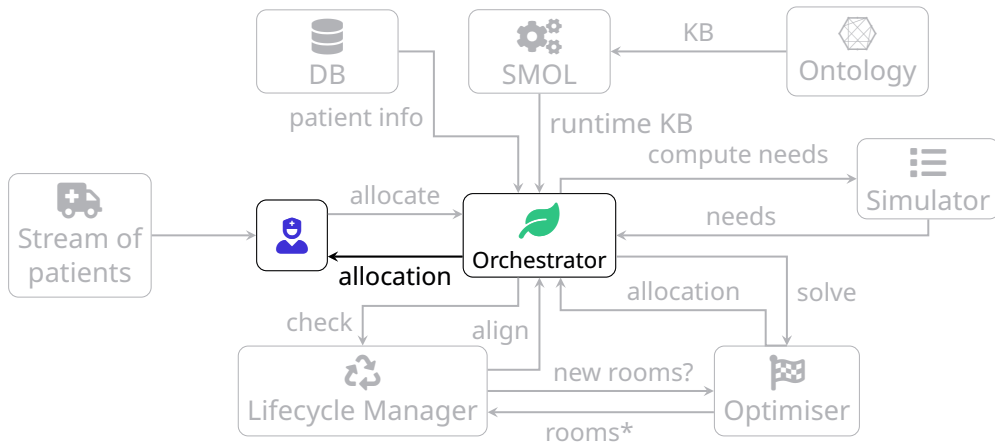
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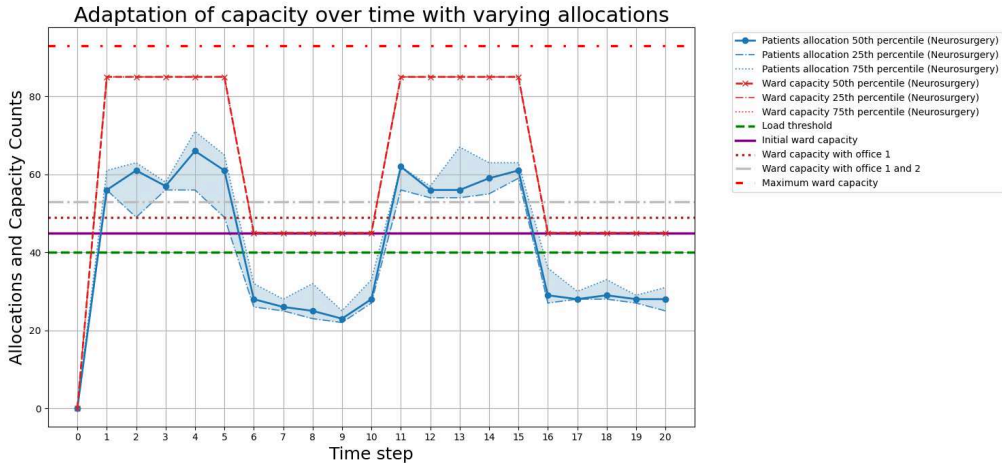
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Allocation of patients



Allocation of patients (cont.)



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- SMOL allows adaptation through semantic lifting, but it comes as a cost
- Stable demand reduces reflection overhead as the semantic lifting is less likely to be triggered
- Penalty-guided optimisation turns the resilience-versus-cost relation into an explicit reasoning element of the architecture
- Current limits: manually tuned penalties and threshold

Future directions

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- Explore different strategies based on uncertainties of non-adaptive approach
- Currently, SMOL provides overhead in the execution, leading to a trade-off between correctness and performance. We plan on investigate and improve the overall performance